hw4

Yifeng Huang

This is the link to the repo: https://github.com/YifengHuang-Bob/Homework4

Problem 1 - Tidyverse: New Zealand

(a) Total Votes by Election Year and Voting Type

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
nzge <- readRDS("data/nzge.rds")
```

```
vote_counts <- nzge %>%
  group_by(election_year, voting_type) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  arrange(desc(total_votes))

print(vote_counts, n = Inf)
```

```
# A tibble: 10 × 3
   election_year voting_type total_votes
           <dbl> <chr>             <dbl>
 1          2014 Party           2416479
```

```
 2          2014 Candidate       2375493
 3          2008 Party           2356536
 4          2008 Candidate       2325598
 5          2005 Party           2286190
 6          2005 Candidate       2260670
 7          2011 Party           2257336
 8          2011 Candidate       2225766
 9          2002 Party           2040248
10          2002 Candidate       2022115
```

(b) Candidate Vote Proportions by Party in 2014

```
candidate_votes_2014 <- nzge %>%
  filter(election_year == 2014, voting_type == "Candidate") %>%
  group_by(party) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  mutate(percent_vote = 100 * total_votes / sum(total_votes)) %>%
  arrange(desc(percent_vote)) %>%
  mutate(percent_vote = round(percent_vote, 2))

print(candidate_votes_2014, n = Inf)
```

```
# A tibble: 25 × 3
   party                            total_votes percent_vote
   <chr>                                  <dbl>        <dbl>
 1 National Party                       1081787        45.5
 2 Labour Party                          801287        33.7
 3 Green Party                           165718         6.98
 4 Conservative Party                     81075         3.41
 5 New Zealand First Party                73384         3.09
 6 Maori Party                            42108         1.77
 7 MANA Movement                          32333         1.36
 8 Informal Candidate Votes               27886         1.17
 9 ACT New Zealand                        27778         1.17
10 United Future                          14722         0.62
11 Aotearoa Legalise Cannabis Party        4936         0.21
12 Internet Party                          4848         0.2
13 Democrats for Social Credit             4647         0.2
14 Ban1080                                 4448         0.19
15 Independent                             3864         0.16
16 NZ Independent Coalition                1929         0.08
```

```
17 Focus New Zealand                        1797         0.08
18 Money Free Party                          391         0.02
19 Communist League                          135         0.01
20 Climate Party                             116         0
21 Human Rights Party                         76         0
22 The Expats                                 70         0
23 Alliance                                   59         0
24 NZ Economic Euthenics Party                51         0
25 Patriotic Revolutionary Front              48         0
```

(c) Winning Parties by Election Year and Voting Type

```
winning_parties <- nzge %>%
  group_by(election_year, voting_type, party) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  group_by(election_year, voting_type) %>%
  slice_max(order_by = total_votes, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  select(election_year, voting_type, party) %>%
  pivot_wider(names_from = voting_type, values_from = party) %>%
  arrange(election_year)

print(winning_parties, n = Inf)
```

```
# A tibble: 5 × 3
  election_year Candidate      Party
          <dbl> <chr>          <chr>
1          2002 Labour Party   Labour Party
2          2005 National Party Labour Party
3          2008 National Party National Party
4          2011 National Party National Party
5          2014 National Party National Party
```

Problem 2 - Tidyverse: Tennis

(a) Number of Tournaments in 2019

```
atp2019 <- read_csv("data/atp_matches_2019.csv", show_col_types = FALSE)
```

```r
tournament_count <- atp2019 %>%
  summarise(n_tournaments = n_distinct(tourney_id))

print(tournament_count)
```

```
# A tibble: 1 × 1
  n_tournaments
          <int>
1           128
```

There were **128 tournaments** in the 2019 ATP Matches dataset, determined by counting the distinct tournament IDs.

(b) Players Winning Multiple Tournaments

```r
multiple_title_winners <- atp2019 %>%
  filter(round == "F") %>%
  group_by(winner_id, winner_name) %>%
  summarise(tournaments_won = n_distinct(tourney_id), .groups = "drop") %>%
  filter(tournaments_won > 1) %>%
  arrange(desc(tournaments_won), winner_name)

print(multiple_title_winners, n = Inf)
```

```
# A tibble: 12 × 3
   winner_id winner_name        tournaments_won
       <dbl> <chr>                        <int>
 1    106233 Dominic Thiem                    5
 2    104925 Novak Djokovic                   5
 3    106421 Daniil Medvedev                  4
 4    104745 Rafael Nadal                     4
 5    103819 Roger Federer                    4
 6    200282 Alex De Minaur                   3
 7    126774 Stefanos Tsitsipas               3
 8    105332 Benoit Paire                     2
 9    106426 Cristian Garin                   2
10    104542 Jo-Wilfried Tsonga               2
11    126610 Matteo Berrettini                2
12    106401 Nick Kyrgios                     2
```

```r
multiple_title_winners %>%
  summarise(
    n_players = n(),
    max_tournaments = max(tournaments_won)
  ) %>%
  print()
```

```
# A tibble: 1 × 2
  n_players max_tournaments
      <int>           <int>
1        12               5
```

Yes. In 2019, **12 players** won more than one tournament. **Dominic Thiem and Novak Djokovic** tied for the most tournament titles, each winning **5 tournaments**.

(c) Comparing Aces Between Winners and Losers

```r
ace_comparison <- atp2019 %>%
  filter(!is.na(w_ace), !is.na(l_ace)) %>%
  mutate(ace_diff = w_ace - l_ace) %>%
  summarise(
    n_matches = n(),
    mean_winner = mean(w_ace),
    mean_loser = mean(l_ace),
    mean_diff = mean(ace_diff),
    se_diff = sd(ace_diff) / sqrt(n())
  ) %>%
  mutate(
    lower_ci = mean_diff - 1.96 * se_diff,
    upper_ci = mean_diff + 1.96 * se_diff
  ) %>%
  select(-se_diff)

print(ace_comparison, width = Inf)
```

```
# A tibble: 1 × 6
  n_matches mean_winner mean_loser mean_diff lower_ci upper_ci
      <int>       <dbl>      <dbl>     <dbl>    <dbl>    <dbl>
1      2694        7.50       5.79      1.70     1.43     1.97
```

Yes. Based on 2,694 matches with complete ace records, winners averaged **7.50 aces** per match, compared with **5.79 aces** for losers. The mean paired difference was approximately **1.70 aces** (the reported means and mean difference were calculated using unrounded values and rounded separately for display), with an approximate 95% confidence interval of **[1.43, 1.97]**.

Since the entire confidence interval is above zero, there is statistical evidence that winners tend to have more aces than losers.

(d) Players with the Highest Win Rate

```
player_results <- bind_rows(
  atp2019 %>%
    transmute(player_id = winner_id, player_name = winner_name, won = 1),
  atp2019 %>%
    transmute(player_id = loser_id, player_name = loser_name, won = 0)
)

top_win_rate <- player_results %>%
  group_by(player_id, player_name) %>%
  summarise(
    matches_played = n(),
    matches_won = sum(won),
    .groups = "drop"
  ) %>%
  filter(matches_played >= 5) %>%
  mutate(win_rate = matches_won / matches_played) %>%
  slice_max(order_by = win_rate, n = 1, with_ties = TRUE) %>%
  arrange(player_name)

print(top_win_rate, n = Inf)
```

```
# A tibble: 1 × 5
  player_id player_name  matches_played matches_won win_rate
      <dbl> <chr>                <int>       <dbl>    <dbl>
1    104745 Rafael Nadal            69          60    0.870
```

Rafael Nadal had the highest win rate among players who participated in at least 5 matches. He won **60 out of 69 matches**, achieving a win rate of approximately **87.0%**.

Problem 3 - Visualization

(a) Major and Minor Spikes in COVID-19 Cases

```
covid <- read_csv("data/us-states.csv", show_col_types = FALSE)
```

```
us_cases <- covid %>%
  group_by(date) %>%
  summarise(
    cases_7day = sum(cases_avg),
    .groups = "drop"
  )
```

```
peak_summary <- us_cases %>%
  mutate(
    wave = case_when(
      date < as.Date("2020-05-16") ~ "Spring 2020",
      date < as.Date("2020-09-16") ~ "Summer 2020",
      date < as.Date("2021-03-15") ~ "Winter 2020-21",
      date < as.Date("2021-07-01") ~ "Spring 2021",
      date < as.Date("2021-11-16") ~ "Summer-Fall 2021",
      date < as.Date("2022-03-16") ~ "Winter 2021-22",
      date < as.Date("2022-10-16") ~ "Summer 2022",
      TRUE ~ "Winter 2022-23"
    )
  ) %>%
  group_by(wave) %>%
  slice_max(cases_7day, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  mutate(
    spike_type = if_else(cases_7day >= 150000, "Major", "Minor")
  ) %>%
  arrange(date)

peak_summary %>%
  select(wave, date, cases_7day, spike_type) %>%
  print(n = Inf)
```

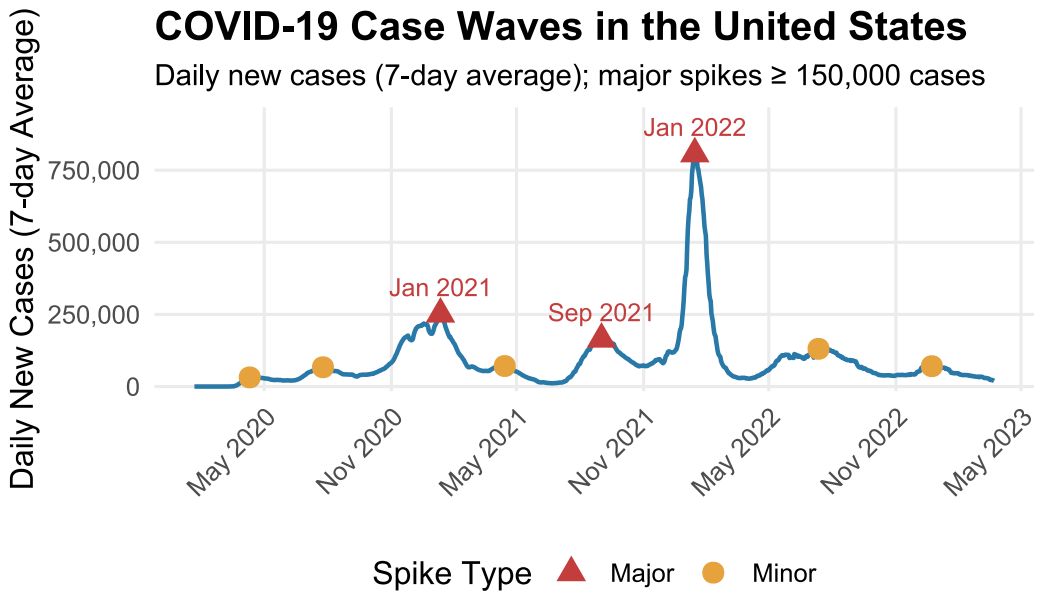
```
# A tibble: 8 × 4
  wave             date       cases_7day spike_type
  <chr>            <date>          <dbl> <chr>
1 Spring 2020      2020-04-10     31700. Minor
```

```
2 Summer 2020      2020-07-25     66787. Minor
3 Winter 2020-21   2021-01-11    251240. Major
4 Spring 2021      2021-04-14     71093. Minor
5 Summer-Fall 2021 2021-09-01    164418. Major
6 Winter 2021-22   2022-01-14    806795. Major
7 Summer 2022      2022-07-12    130670. Minor
8 Winter 2022-23   2022-12-23     70512. Minor
```

```
invisible(Sys.setlocale("LC_TIME", "C"))

ggplot(us_cases, aes(x = date, y = cases_7day)) +
  geom_line(color = "#2878A8", linewidth = 0.8) +
  geom_point(
    data = peak_summary,
    aes(color = spike_type, shape = spike_type),
    size = 3.5
  ) +
  geom_text(
    data = peak_summary %>% filter(spike_type == "Major"),
    aes(label = format(date, "%b %Y")),
    color = "#C43D3D",
    vjust = -1,
    size = 3.3
  ) +
  scale_color_manual(
    values = c("Major" = "#C43D3D", "Minor" = "#E6A23C")
  ) +
  scale_shape_manual(
    values = c("Major" = 17, "Minor" = 16)
  ) +
  scale_x_date(
    date_breaks = "6 months",
    date_labels = "%b %Y"
  ) +
  scale_y_continuous(
    labels = scales::label_comma(),
    expand = expansion(mult = c(0.02, 0.20))
  ) +
  labs(
    title = "COVID-19 Case Waves in the United States",
    subtitle = "Daily new cases (7-day average); major spikes ≥ 150,000 cases",
    x = NULL,
```

```
    y = "Daily New Cases (7-day Average)",
    color = "Spike Type",
    shape = "Spike Type",
     caption = "Source: The New York Times | 50 states, D.C., and five U.S.
territories"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 15),
    plot.subtitle = element_text(size = 11),
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid.minor = element_blank(),
    legend.position = "bottom",
    plot.caption = element_text(hjust = 0)
  )
```



Based on the 7-day rolling average of daily new COVID-19 cases, we identified **8 distinct waves**, consisting of **3 major spikes and 5 minor spikes**.

We classified a spike as major if its peak reached at least 150,000 daily cases, and minor otherwise. The three major spikes occurred in **January 2021, September 2021, and January 2022**, with the largest peak reaching approximately **806,795 daily cases** in January 2022.

The five minor spikes occurred in **April 2020, July 2020, April 2021, July 2022, and December 2022**. These classifications are based on visually identified epidemic waves and an exploratory threshold, rather than a formal peak-detection procedure.

(b) Comparing States with the Highest and Lowest Case Rates

```
state_rates <- covid %>%
  filter(!state %in% c(
    "American Samoa", "District of Columbia", "Guam",
    "Northern Mariana Islands", "Puerto Rico", "Virgin Islands"
  )) %>%
  group_by(state) %>%
  summarise(
    overall_rate_per_100k = sum(cases_avg_per_100k),
    .groups = "drop"
  )

extreme_states <- bind_rows(
  state_rates %>%
    slice_max(overall_rate_per_100k, n = 1, with_ties = TRUE) %>%
    mutate(category = "Highest"),
  state_rates %>%
    slice_min(overall_rate_per_100k, n = 1, with_ties = TRUE) %>%
    mutate(category = "Lowest")
) %>%
  select(category, state, overall_rate_per_100k)

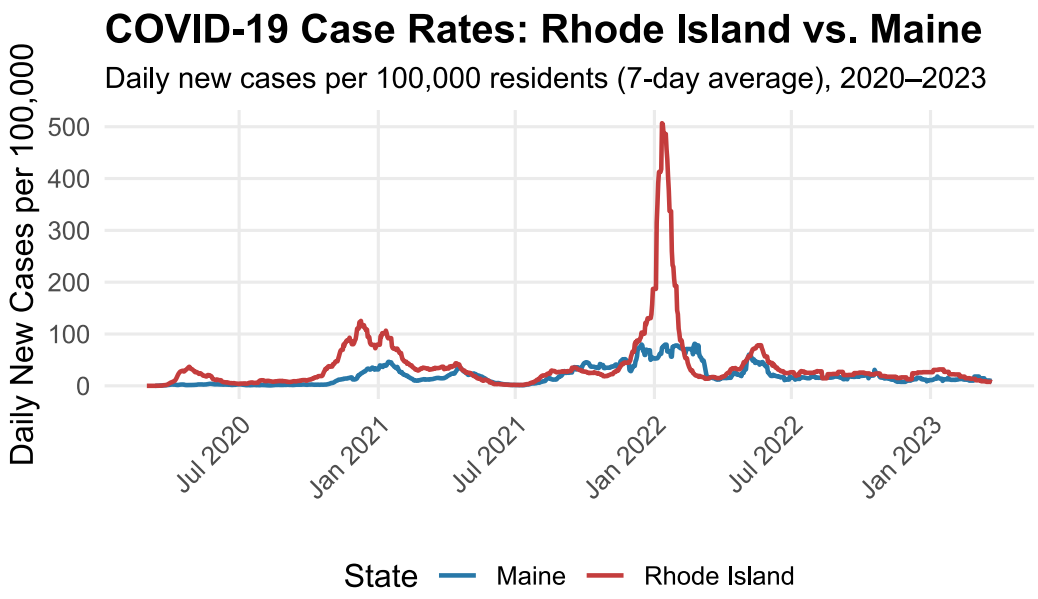
print(extreme_states, n = Inf)
```

```
# A tibble: 2 × 3
  category state        overall_rate_per_100k
  <chr>    <chr>                        <dbl>
1 Highest  Rhode Island                43296.
2 Lowest   Maine                       22020.
```

```
state_comparison <- covid %>%
  filter(state %in% extreme_states$state)
```

```
ggplot(
  state_comparison,
  aes(x = date, y = cases_avg_per_100k, color = state)
```

```
) +
  geom_line(linewidth = 0.8) +
  scale_color_manual(
    values = c(
      "Rhode Island" = "#C43D3D",
      "Maine" = "#2878A8"
    )
  ) +
  scale_x_date(
    date_breaks = "6 months",
    date_labels = "%b %Y"
  ) +
  scale_y_continuous(
    labels = scales::label_comma()
  ) +
  labs(
    title = "COVID-19 Case Rates: Rhode Island vs. Maine",
   subtitle = "Daily new cases per 100,000 residents (7-day average), 2020–2023",
    x = NULL,
    y = "Daily New Cases per 100,000",
    color = "State",
    caption = "Source: The New York Times | Rhode Island and Maine"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 15),
    plot.subtitle = element_text(size = 11),
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid.minor = element_blank(),
    legend.position = "bottom",
    plot.caption = element_text(hjust = 0)
  )
```



Source: The New York Times | Rhode Island and Maine

Among the 50 U.S. states, **Rhode Island** had the highest approximate cumulative COVID-19 case rate, with **43,296 cases per 100,000 residents**, while **Maine** had the lowest, with **22,020 cases per 100,000 residents**. Rhode Island's overall rate was approximately twice that of Maine. These cumulative rates were estimated by summing the daily 7-day rolling average case rates per 100,000 residents.

The trajectories show several important differences. **Rhode Island experienced earlier and substantially sharper spikes**, particularly during the winter of 2020–2021 and January 2022. Its January 2022 peak exceeded **500 daily cases per 100,000 residents**, far higher than Maine's rate during the same period.

In contrast, **Maine generally experienced lower and more gradual increases**, with some peaks occurring later than those in Rhode Island. Although Rhode Island had the higher overall rate, Maine occasionally reached comparable or higher daily case rates during certain periods.

Overall, Rhode Island experienced more pronounced fluctuations and substantially higher peaks, whereas Maine's trajectory was generally less extreme.

(c) First Five States to Experience Substantial COVID-19 Spread

```
threshold_comparison <- covid %>%
  filter(
    state %in% state_rates$state,
    !is.na(cases_avg_per_100k)
  ) %>%
```

```
  crossing(threshold = c(3, 5, 10)) %>%
  filter(cases_avg_per_100k >= threshold) %>%
  group_by(threshold, state) %>%
  summarise(
    first_date = min(date),
    .groups = "drop"
  ) %>%
  group_by(threshold) %>%
  slice_min(first_date, n = 5, with_ties = TRUE) %>%
  ungroup() %>%
  arrange(threshold, first_date, state)

print(threshold_comparison, n = Inf)
```

```
# A tibble: 17 × 3
   threshold state         first_date
       <dbl> <chr>         <date>
 1         3 New York      2020-03-20
 2         3 Louisiana     2020-03-23
 3         3 New Jersey    2020-03-23
 4         3 Connecticut   2020-03-25
 5         3 Massachusetts 2020-03-25
 6         3 Michigan      2020-03-25
 7         5 New York      2020-03-21
 8         5 New Jersey    2020-03-24
 9         5 Louisiana     2020-03-26
10         5 Massachusetts 2020-03-27
11         5 Connecticut   2020-03-28
12         5 Michigan      2020-03-28
13        10 New York      2020-03-22
14        10 New Jersey    2020-03-27
15        10 Massachusetts 2020-03-30
16        10 Connecticut   2020-03-31
17        10 Louisiana     2020-03-31
```

```
first_five_states <- threshold_comparison %>%
  filter(threshold == 10) %>%
  select(state, first_date)

print(first_five_states, n = Inf)
```

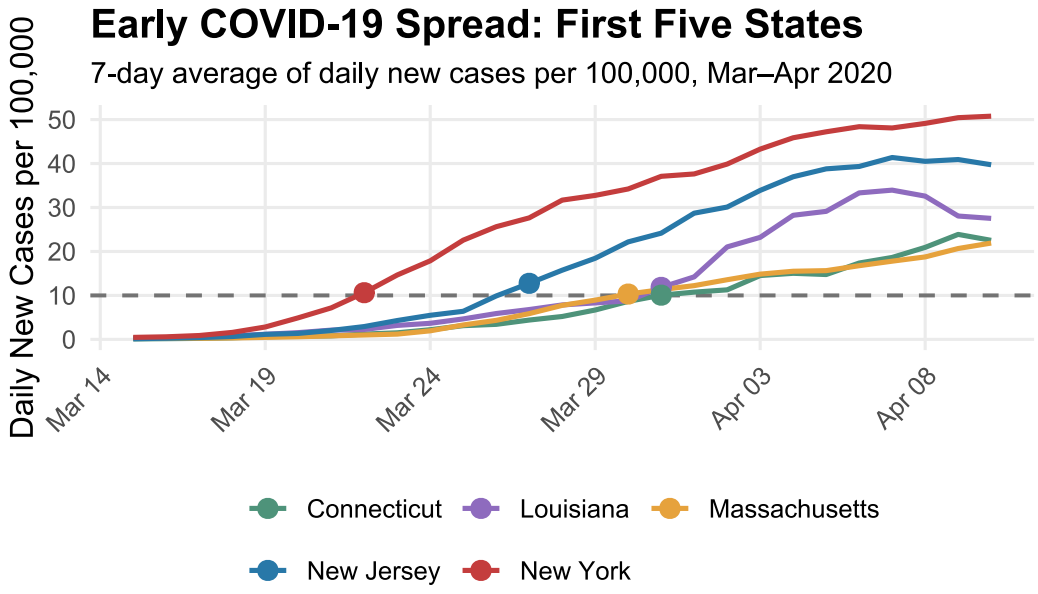
```
# A tibble: 5 × 2
  state         first_date
  <chr>         <date>
1 New York      2020-03-22
2 New Jersey    2020-03-27
3 Massachusetts 2020-03-30
4 Connecticut   2020-03-31
5 Louisiana     2020-03-31
```

```
early_trajectories <- covid %>%
  filter(
    state %in% first_five_states$state,
    date >= as.Date("2020-03-15"),
    date <= as.Date("2020-04-10")
  )

threshold_points <- covid %>%
  inner_join(first_five_states, by = "state") %>%
  filter(date == first_date)
```

```
ggplot(
  early_trajectories,
  aes(x = date, y = cases_avg_per_100k, color = state)
) +
  geom_hline(
    yintercept = 10,
    linetype = "dashed",
    color = "gray45",
    linewidth = 0.7
  ) +
  geom_line(linewidth = 0.9) +
  geom_point(
    data = threshold_points,
    aes(x = first_date, y = cases_avg_per_100k, color = state),
    inherit.aes = FALSE,
    size = 3
  ) +
  scale_color_manual(
    values = c(
      "New York" = "#C43D3D",
      "New Jersey" = "#2878A8",
      "Massachusetts" = "#E6A23C",
```

```
    "Connecticut" = "#4E937A",
    "Louisiana" = "#8E6BBE"
  )
) +
guides(
  color = guide_legend(nrow = 2, byrow = TRUE)
) +
scale_x_date(
  date_breaks = "5 days",
  date_labels = "%b %d"
) +
scale_y_continuous(
  labels = scales::label_comma()
) +
labs(
  title = "Early COVID-19 Spread: First Five States",
  subtitle = "7-day average of daily new cases per 100,000, Mar–Apr 2020",
  x = NULL,
  y = "Daily New Cases per 100,000",
  color = NULL,
  caption = "Source: The New York Times | Dashed line: 10 cases per 100,000"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", size = 15),
  plot.subtitle = element_text(size = 11),
  axis.text.x = element_text(angle = 45, hjust = 1),
  panel.grid.minor = element_blank(),
  legend.position = "bottom",
  plot.caption = element_text(hjust = 0)
)
```



We defined substantial COVID-19 spread as reaching at least **10 daily new cases per 100,000 residents**, based on the 7-day rolling average. Under this criterion, the first five states were **New York (March 22, 2020), New Jersey (March 27), Massachusetts (March 30), Connecticut (March 31), and Louisiana (March 31)**.

The plot shows that **New York experienced the earliest and most rapid increase**, followed by New Jersey. Both states exhibited substantially higher case rates by early April 2020. Louisiana also experienced relatively rapid growth, while Massachusetts and Connecticut showed more gradual increases during the same period.

To assess the sensitivity of our findings, we also considered thresholds of **3 and 5 daily cases per 100,000 residents**. The same five states appeared among the earliest states under all three thresholds, although Michigan was also included at the lower thresholds due to ties.

These findings are based on an exploratory threshold rather than a formal statistical test. The identified onset dates depend on the chosen threshold and reported case data, which may be affected by differences in testing and reporting across states.